



Normative sample of the PEAK relational training system: Direct training module and subsequent comparisons to individuals with autism



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ABSTRACT

The present data provide a normative sample of the PEAK: direct training module assessment and a subsequent comparison to individuals with autism. Altogether, 206 typically developing participants and 94 participants with autism took part in the study. For the normative sample, there was a strong relationship between PEAK total score and age ($r = .659, p < .01$), and a cubic regression provided a strong fit for the data ($R^2 = .821, t = 18.51, p < .01$). The results from the autism sample suggest that there was no significant correlation between PEAK total score and age ($r = .021, p = .861$), and that PEAK total scores for the autism group were significantly lower than the normative sample ($t(275) = 10.63, p < .001$). The data suggest that PEAK may be especially useful as an assessment and curriculum guide for individuals with autism, and future research should be conducted on the increasingly complex topographies of human language and cognition that PEAK affords clinicians.

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1. Introduction

Promoting the emergence of advanced knowledge relational training system (PEAK; Dixon, 2014a, 2014b) is an assessment and curriculum guide designed to promote the cognitive and language development of individuals with autism. PEAK takes a behavior analytic approach to understanding language and cognition, and posits that by understanding environmental variables that promote learning, teaching strategies can begin to effectively promote language acquisition. Through adherence to this philosophy, behavior analytic approaches to language development have been demonstrably more effective than many alternative approaches (Green, 1996; Smith, 1996), and have expedited the transition of children with autism into regular education classrooms of an appropriate age level (e.g., Fenske, Zalsenski, Krantz, & McClannahan, 1985; Lovaas, 1987). Not only have behavior analytic approaches been shown to improve language, but they have also been shown to increase the overall intelligence and cognitive functioning of children with autism (Peters-Scheffer, Didden, Korzilius, & Sturmey, 2011).

PEAK is comprised of four modules: direct training, generalization, equivalence, and transformation. The focus of the present paper is on the direct training module, which has been noted by Dixon (2014a, 2014b) as containing the most basic skill set of PEAK. The direct training model is based on the simple and complex verbal relations proposed by Skinner (1957), and is largely taught through discrete trial training. Discrete trial training involves the presentation of a discriminative

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stimulus, such as “which is a ball?”; a learner response, such as pointing to a ball instead of distracter stimuli; reinforcement for correct responding; and ensuring short intervals between discrete trials (Brown-Chidsey & Steege, 2004; Koegel, Russo, & Rincover, 1977). Direct training procedures have been the basis of similar assessment programs, such as VB-MAPP (Sundberg, 2008) and ABLIS-R (Partington, 2002), and serve as the empirically grounded foundation of the PEAK curriculum. The remaining PEAK modules build upon the directly trained skills established through the PEAK direct training module, and taken together; these four modules have been noted to represent the apex of the science’s current understanding of language development.

As an assessment tool, the PEAK direct training module has been gathering support as a psychometrically sound assessment of the functioning of individuals with disabilities. Dixon, Carman, et al. (in press-a) and Dixon, Whiting, Rowsey and Belisle (in press-b) assessed the convergent validity of the PEAK by investigating its relationship with both the Peabody Picture Vocabulary Test (PPVT) and the Illinois Common Core Learning Standards (ICCLS) with individuals with autism and other disabilities. These tools were selected because they are both measures of children’s language ability, which is a primary focus of PEAK. The results suggested a strong positive relationship between PEAK, PPVT, and ICCLS. In addition, Dixon et al. demonstrated high interrater reliability by comparing the PEAK scores obtained by certified behavior analysts and novice assessors. Dixon, Carman, et al. (in press-a) and Dixon, Whiting, et al. (in press-b) expanded on the results of Dixon et al. by suggesting that PEAK is also a valid assessment of cognitive ability. Dixon, Whiting, Rowsey, and Belisle assessed the relationship between PEAK, age, and intelligence in a sample of children and adolescents with autism. The results suggested strong convergent validity between PEAK and IQ scores, and there was no relationship between PEAK and age or age and IQ. Tohidian and Tabatabaie (2010) propose that language is at the center of human cognitive development, and preliminary research suggests that the PEAK direct training module is a valid and reliable behavioral measure of both these constructs.

Although initial psychometric properties of PEAK have been promising, a normative sample could strengthen this growing body of support. Normative data accomplishes two essential purposes (Gregory, 2011): First, normative data allows for an analysis of typical development in its own right. Second, normative data allows for a point of comparison to identify sources of abnormality. One group of psychometric tools that have been used to measure intellectual or cognitive ability are the Wechsler tests, which are the most frequently used intelligence tests with children with autism (Oliveras-Rentas, Kenworthy, Roberson, Martin, & Wallace, 2011). Normalization of the WISC-IV was conducted for the purposes of standardization (Gregory). Altogether, 2200 participants comprised the normative sample to ensure that the standardization was based on a representative sample (Wechsler, 2003). By allowing for a frame of reference, the WISC-IV normative sample now allows for differential diagnoses and treatment decisions to be made based on the degree to which children deviate from typically developing children of the same age (Goldstein et al., 2008), and for monitoring relative progress (Koegel, Koegel, & Smith, 1997). From a behavioral approach to language development, the ABLIS and the VB-MAPP have also provided samples of normative data. Partington, Bailey, and Pritchard (2010) provided normative data for the ABLIS, suggesting that most typically developing children mastered most of the ABLIS items by 5-years of age. Sundberg and Sundberg (2011) provided normative data for the intraverbal subsection of the VB-MAPP for comparison to children with autism. These data suggested that children could emit basic intraverbal responses by 31-months, and could make verbal discriminations by 3-years of age. These data suggest that normalization data have utility as psychometric additions to behavioral measures of language; however, the results of these studies also suggest that only the most basic language repertoires are understood with our current technologies.

In line with these data, Dymond, O’Hora, Whelan, and O’Donovan (2006) found that much of the current behavioral literature on language acquisition has been conceptual rather than empirical, and according to Dixon, Small, and Rosales (2007), the empirical literature has only truly analyzed these basic language processes. Language is a primary deficit experienced by individuals with autism, and therefore conceptual discussions and empirical literature with a limited scope are not sufficient for understanding and improving the lives of individuals with this disorder. Therefore, the purpose of the present study was twofold. First, the present study developed a normative sample of the PEAK direct training module. Second, the present study compared a sample of individuals with autism to the normative sample. By developing a normative sample, specific cognitive and language deficits experienced by individuals with autism can be identified as they deviate from typical cognitive and language development. Once indicated, programming designed to teach these skills can begin to ‘bridge the gap’ between the skill repertoires of individuals with autism and their typically developing peers.

2. Experiment 1: developing a normative sample of PEAK

2.1. Materials and methods

2.1.1. Participants

A total of 206 individuals (113 females, 93 males) took part in the present experiment. All of the participants were enrolled in regular education classrooms, and none of the participants had a diagnosis of autism. The participants were recruited by graduate students at Southern Illinois University based on a priori relationships. The age of the participants ranged from 1-year to 21-years of age ($M = 10.06$, $SD = 6.08$). The sample was relatively homogenous, and incorporated participants who were White (88%), Black (7%), Hispanic (3%), and Other (2%). In addition, participants were assessed across 19 American states and 1 Canadian province. Table 2 contains identification of the numbers of subjects per age.

2.1.2. Materials

The PEAK direct training assessment was used as a measure of the language and cognitive skills of each participant. The assessment is comprised of 184 items, or skills, that are typically directly taught to children with- and without- disabilities. The skills assessed include prerequisite learning skills (e.g., eye contact and keeping hands still), vocal skills (e.g., imitating sounds and imitating words), writing skills (e.g., copying text and transcribing speech), conversational skills (e.g., telling jokes and answering questions), basic and advanced math skills (e.g., addition and subtraction), and advanced conversational skills (e.g., identifying metaphors and communicating uncertainty). Each of the 184 items that comprise PEAK direct training module are listed on the assessment, with check boxes to indicate whether each item is in the participants' repertoire. The time to complete each assessment ranged from 15- to 65-min. The disparity in time to complete the assessment depended on the amount of items that needed to be directly tested, based on the skill repertoire of the participant and the familiarity of the assessor.

2.1.3. Procedure

PEAK assessments were completed by trained graduate students in behavior analysis who were familiar with the participants being assessed. Dixon (2014a) suggested that assessor familiarity might yield a more accurate PEAK assessment over a naïve assessor. Assessors evaluated the participants by first indicating items on the assessment that they have observed the participant performing, or know that they are able to perform independently. Once this was completed, the assessors tested skills in question. For example, if an assessor did not know if a participant was able to receptively label animals, they would present the participant with a picture of an animal and distracter stimuli, and asked "which is an animal?". If participants were able to perform the skill, then the assessor indicated on the assessment that the item was mastered. If participants were unable to perform the skill, then the assessor indicated on the assessment that the item was not mastered. In situations where assessors did not indicate if the skill was present or absent in the participant's repertoire (i.e. failed to test the item), then the skill was assumed to not be mastered and was indicated as much on the assessment. The total items mastered were then aggregated to arrive at a PEAK total score, which has been used as a quantifiable measure of children's directly trained behavior repertoire in the PEAK literature.

2.1.4. Data analyses

2.1.4.1. The relationship between PEAK and age. Several statistical and visual analyses were conducted to evaluate the relationship between the participants' PEAK total score and age. First, Pearson correlation coefficients were conducted to determine the relationship between PEAK total score, age, and sex. The primary relationship of interest was PEAK total score and age; however, the correlation between both of these variables and sex was assessed to determine if sex was a moderating factor. Second, a scatterplot was created to visually assess the nature of the relationship between PEAK total score and age. Regression analyses were then conducted to fit several functions to the scatterplot data in to determine the trajectory of skill acquisition over time. The regression analyses were compared by their goodness of fit, as well as their significance. Finally, a one-way ANOVA was conducted to determine if the differences between PEAK total scores at each age group were truly significant ($p < .05$), based on the best-fit regression.

2.1.4.2. PEAK distribution within age groups. To compare the distribution within age groups, the participants were divided into groups of 2-years based on their age (i.e. 1–2, 3–4, 5–6 . . . 19–21). The participants were grouped in this manner to increase the number of participants in each group for statistical evaluation. The 19–21 group had three different ages to ensure that the number of participants in this group was similar to the other groups. Mean PEAK total score, standard deviation, kurtosis, and skewness were determined for each group to determine the ages at which the PEAK direct training module serves as an appropriate assessment of language and cognition.

2.1.4.3. PEAK item development across age groups. To evaluate the order in which specific skills develop in a typically developing population, an item analysis across age groups was conducted. The same grouping criterion that was used to determine the distribution within age groups was used in the present analysis (e.g., 1–2, 2–3, etc.). For each age group, the percentage of participants who mastered each item was assessed. Then, the age at which greater than 80% of the participants were able to complete each skill was determined, and each item was categorized based on the earliest age where greater than 80% of the participants mastered the skill. The criteria of 80% was selected because it ensured that outliers did not affect the analysis, while being stringent enough to provide sufficient variability to allow for grouping. By providing a description of the age at which greater than 80% of participants mastered each skill, consumers may compare their participants' and clients' repertoire to that of average typically developing individuals of the same age group and pinpoint specific skill deficits.

2.2. Results

2.2.1. The relationship between PEAK and age

A Pearson correlational analysis was conducted to determine the relationship between PEAK total score, age, and sex. The mean PEAK total score was 156.2 (SD = 48.8) and the mean age was 10.0 (SD = 6.0). The results of the Pearson correlational analysis suggested that there was a strong positive relationship between PEAK total score and age ($r = .659$, $p < .01$). Also, the

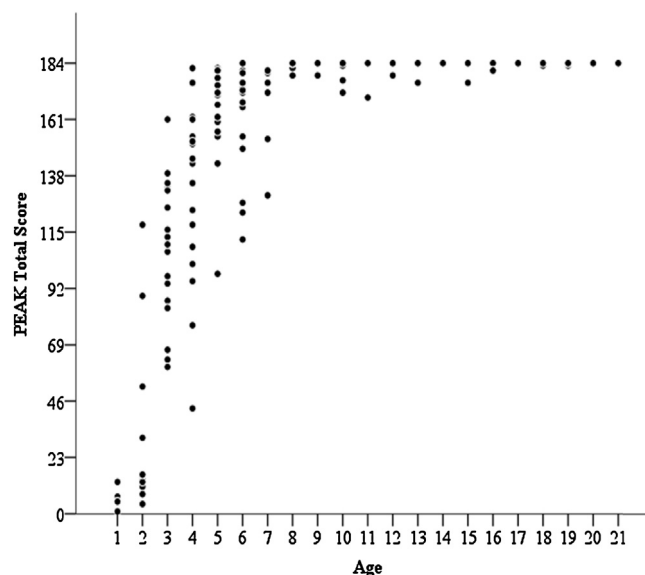


Fig. 1. The relationship between PEAK total score and age for the normative sample.

results suggest that there was no correlation between PEAK total scores and gender ($r = .014$, $p = .837$), or between age and sex ($r = -.017$, $p = .807$), suggesting that gender was not a mediating factor. A scatterplot displaying the relationship between PEAK total scores and age is displayed in Fig. 1. Visual analysis of the scatterplot confirms that there was a positive correlation between PEAK total score and age. As age increased, PEAK total scores correspondingly increased until approximately the age of 8, at which point there was a ceiling effect. At around the age of 8, most participants were able to perform all 184 items of the PEAK direct training module.

Several regression lines, both linear and nonlinear, were fit to the scatterplot to determine the trajectory of the participants' directly trained skill repertoire. A summary of the regression lines, their associated R -square values, and significance of fit are summarized in Table 1. A linear regression produced an R -square of .435, and produced a significant fit to the data ($p < .001$). Although the fit was significant, the R -square was relatively low, so additional nonlinear regression lines were fit to the data, including: Logarithmic, quadratic, cubic, and logistic. The cubic regression produced the strongest fit. The R -square for the cubic equation was .821 and produced a significant fit ($p < .01$). Visual analysis of the relationship between PEAK and age also suggest that the development of a directly trained skill repertoire is not linear. An ANOVA was conducted based off of the cubic regression, and the results suggested that the differences between ages in terms of PEAK total score was significant ($F(3, 204) = 311.277$, $p < .001$).

2.2.2. PEAK distribution within age groups

The results from the PEAK distribution within age groups are summarized in Fig. 2. The boxplot displays the mean, standard deviation, standard error, and outliers associated with each age group. A horizontal line inside each box indicates the means for each group. The boxes represent one standard deviation from the mean, the error bars represent the standard error, and the asterisks represent outliers in the data. Visual inspection of the means suggest that, as age increases, so does the mean PEAK total score, and there appears to be a ceiling effect around the age of 9. There also appeared to be a greater degree of variability in the data associated with the 1–2 and 3–4 age groups. As age increases, the variability in the data appears to decrease around a ceiling of 184 items, which was the maximum score that participants could achieve.

The means, standard deviations, kurtoses, and skewness associated with each group are summarized in Table 2. The lowest mean was associated with the 1–2 group ($M = 24.93$, $SD = 33.46$), and the highest standard deviation was associated with the 3–4 group ($M = 117.94$, $SD = 34.32$). Conversely, the highest mean and lowest standard deviation were both

Table 1

A summary of regression lines and their goodness of fit when applied to the relationship between PEAK total score and age.

Regression type	Total cases	R	R -square	Standard error of estimate	B	t	Sig.
Linear	206	.659	.435	36.780	5.29	12.585	<.01
Logarithmic	206	.829	.688	27.337	53.08	21.301	<.01
Quadratic	206	.840	.705	26.624	23.40	17.275	<.01
Cubic	206	.906	.821	20.816	55.65	18.510	<.01
Logistic	206	.790	.624	1.553	.72	56.332	<.01

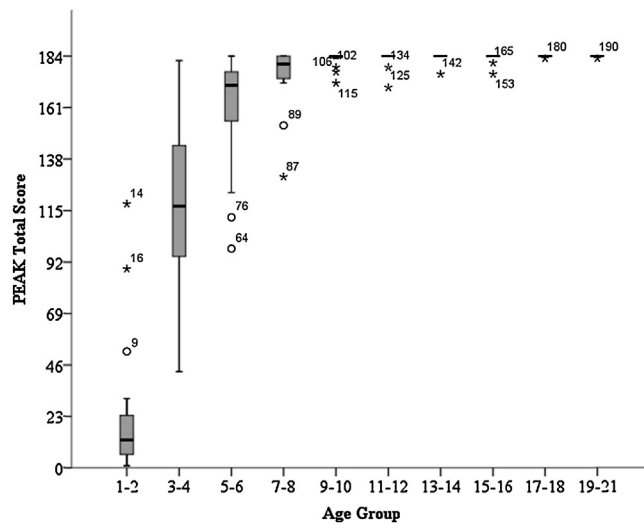


Fig. 2. Box-plot of PEAK total score by age group. Horizontal lines represent the mean PEAK total score, boxes represent 1 standard deviation, whiskers represent 1 standard error, circles represent scores that fall 2 standard deviations from the mean, and asterisks represent scores that fall 3 standard deviations from the mean.

associated with the 17–18 group ($M = 183.93$, $SD = .25$). The kurtoses of the data ranged from $-.57$ to 16.000 ; however, extreme high kurtoses only occurred in the higher age groups. Kurtoses between the ages of 1–8 ranged from $-.579$ to 4.72 . Scores that are below $2 \pm$ represent a generally normal distribution. Extreme high kurtoses suggest that scores are clustered at one tail or the other, so the ceiling effect following the age of 8 may explain the high kurtoses between the ages of 9–21 (range, 7.28 – 16.00). All of the data, except for the 1–2 group, were negatively skewed (range, -4.00 to 2.09). Again, scores that are below $2 \pm$ represent a generally normal distribution, and scores above this cutoff suggest that the means are problematically affected by outliers. Analogous to the kurtoses of the present data, skewness between the ages of 1–8 ranged from -2.22 to 2.09 ; whereas skewness between the ages of 9–21 ranged from -2.72 to -4.00 .

2.2.3. PEAK item analysis across age groups

The cumulative number of skills acquired at each age group is summarized in Fig. 3, and a list of the items acquired at each age group is provided in Table 3. Between the ages of 1–2, only two items were mastered by greater than 80% of the participants. These items were eye contact and modeled play, which are considered prerequisite learning skills. The greatest percentage increase of items mastered occurred between 3 and 4. A total of 65 items were mastered, which was a 3250% increase from the 1–2 age group. By the age of four, participants were able to hold eye contact, vocally and physically imitate models, and vocally identify various stimuli. The greatest overall increase in the number of items mastered occurred between 5 and 6. A total of 78 items were mastered, which was a 116% increase from the cumulative number of items mastered by the previous age groups. By the age of six, participants increased the number of stimuli that they could vocally identify, as well as acquired the ability to engage in simple vocal tasks such as completing the alphabet, following multiple step directions, making guesses about the future, and expressing simple emotions. Between 7 and 8, there was an increase of 27 items mastered, which was an increase of 18%. By the age of eight, greater than 80% of the participants had mastered 94% of the PEAK direct training items. Many had acquired the ability to self-edit their speech, vocally identify their own

Table 2

The mean, standard deviation, skewness, and kurtosis for PEAK total scores across age groups.

Age group	<i>n</i>	Mean	SD	Kurtosis	SE of kurtosis	Skewness	SE of skewness
1–2	16	24.93	33.46	3.72	1.09	2.09	.56
3–4	36	117.94	34.32	-.57	.76	-.18	.39
5–6	31	162.74	21.79	2.02	.82	-1.57	.42
7–8	12	174.08	16.40	4.72	1.23	-2.22	.63
9–10	20	182.70	3.13	7.28	.99	-2.72	.51
11–12	19	183.00	3.34	14.19	1.01	-3.70	.52
13–14	16	183.50	2.00	16.00	1.09	-4.00	.56
15–16	15	183.26	2.15	10.62	1.12	-3.22	.58
17–18	16	183.93	.25	16.00	1.09	-4.00	.56
19–21	11	183.90	.30	11.00	1.27	-3.31	.66

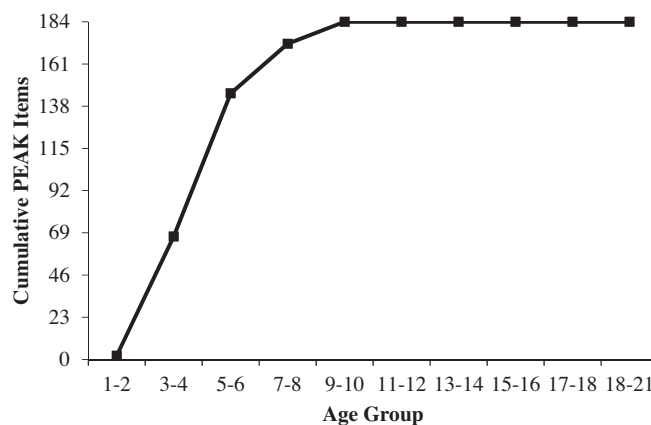


Fig. 3. The cumulative number of PEAK items where >80% of participants demonstrated mastery of the skill across age groups.

behaviors, and tell jokes with proper delivery. The remaining 12 items were acquired between the ages of 9 and 10, which was an increase of 6%. By the age of ten, participants were able to tell time, solve advanced mathematical problems, and tell jokes to an audience with proper delivery. For a complete list of the PEAK items for reference to Table 3, refer to the PEAK Direct Training Assessment (Dixon, 2014a, 2014b).

2.3. Discussion

The present data provide an initial normalization of the PEAK direct training module for comparison to children with autism and other disabilities. By providing a normative sample, the present study allows researchers and clinicians to compare participants and clients to typically developing peers of the same age group. To understand abnormal language and cognitive development, it is crucial to develop a detailed understanding of normal language and cognitive development. The present data suggest that direct training items are related to age in a lawful manner. The cubic nature of this development suggest that directly trained skills are developed rapidly early in life, and the rate of increase declines as children age. If our sample had only contained children under the age of 9, the distribution of scores would have been linearly related. It was the inclusion of the additional ages groups which resulted in the cubic distribution. The source of this decrease in acquisition rate, however, is not well understood. One possible explanation is that the decrease in rate is the result of a ceiling effect at approximately the age of eight. Although children continue to acquire directly trained skills, the directly trained skills past this age are outside of the ambit of the PEAK direct training module. Another potential explanation is that the decrease in rate could be a function of other forms of learning, such as generalization and equivalence, becoming more salient as children enter adolescence and adulthood. It is conceivable that as prerequisite skills are directly trained, advanced skills begin to emerge.

The psychometric properties of the PEAK direct training module, such as skewness and kurtosis, suggest that the PEAK direct training module is a sound assessment of children's behavior repertoire to the point of a typically developing 8-year-old. As such, the normative data may serve as an advantageous point of comparison for individuals with disabilities

Table 3

The PEAK direct assessment items that were completed by >80% of the members in each age group.

Age group														
1-2			3-4			5-6			7-8			9-10		
1A	1B	5D	7K	9C	13G	6C	9H	11G	12M	13M	14V	9E	14A	10H
2B	2A	5E	8A	9D	13Q	6D	9I	11H	12Q	13N	14W	9N	14B	10L
	3A	5F	8B	9J	13R	6G	9O	11J	12R	13O	14X	10B	14C	10R
	3B	5G	8C	9K	13T	6H	9P	11K	12S	13S		10F	14D	11B
	3C	5H	8D	9L	13X	6I	10A	11L	12T	13U		10Q	14F	11D
	3D	6A	8E	9M		6J	10G	11P	13A	13V		11I	14K	11M
	4A	6B	8F	10C		7G	10I	11Q	13B	13W		11T	14L	11O
	4B	6E	8I	10D		7H	10J	11R	13C	14E		12H	14O	12B
	4C	6F	8J	10E		7I	10K	11S	13E	14G		12I	14Q	12V
	4D	7A	8K	10M		7J	10N	12C	13F	14H		12J	14R	14N
	4E	7B	8L	10O		7L	10P	12D	13H	14I		12K	14T	14P
	4F	7C	8M	11N		8G	11A	12E	13I	14J		12N	14Y	14Z
	5A	7D	8N	12A		8H	11C	12F	13J	14M		12O		
	5B	7E	9A	12P		9F	11E	12G	13K	14S		12U		
	5C	7F	9B	13D		9G	11F	12L	13L	14U		13P		

who function at a level lower than a typically developing 8-year-old. Beyond this point, more advanced assessments such as the PEAK generalization module, may be needed to assess language and cognitive development.

The item by age comparison may have clinical utility, as consumers of the PEAK direct training modules can indicate specific items that individuals lack relative to their typically developing peers. In this way, consumers could align the clients that they serve with a typically developing language and cognitive trajectory. This would be accomplished by completing the PEAK direct training assessment with a learner, and targeting items that members of the same age group have typically mastered. The results suggest that it may be chronologically inappropriate to teach a 3-year-old to understand metaphorical emotions; however, this skill may be appropriate for an 8- or 9-year-old. Taken together, these data provide a normative sample to which individuals with autism and other disabilities can be compared. In addition, these data suggest that the PEAK direct training module assesses language and cognitive development beyond current behavioral measures, such as ABLLS and VB-MAPP, and may provide a superior understanding of typical language development in its own right. Although the results are beneficial for the purposes of normalization, the data must also be compared to individuals with autism. Language and cognitive skills are a primary deficit experienced by individuals with autism, and understanding the source of these deficits is imperative for guiding assessment and treatment of the disorder.

3. Experiment 2: comparing students with autism to the normative sample

3.1. Materials and methods

3.1.1. Participants

A total of 94 students (19 females, 75 males), enrolled in a Midwestern American school for children with autism, participated in the present study. Of the 94 students, 76% had a diagnosis of autism, and the remaining 24% had other diagnosed developmental or intellectual disabilities. The students who had a diagnosis other than autism were excluded from the analyses to ensure the homogeneity of the sample. As such, 70 students (12 females, 58 males) remained in the study, and the age of the participants ranged from 5-years to 22-years of age ($M = 14.3$, $SD = 4.4$).

3.1.2. Materials and procedure

As in Study 1, the PEAK direct training assessment was used to evaluate the directly trained skill repertoire of children with autism. Graduate students in behavior analysis, who had worked with the participants previously, completed the assessments. The assessment for each participant was completed in a single session; if there were problem behaviors during the assessment, then the assessment was discontinued and reinitiated at a later date. The assessor began by indicating items that they have observed the participant doing in the past, and then directly assessed items that were in question. Because of the diversity of the participants' skills, the assessments took 10–120 min to complete.

3.1.3. Data analyses

3.1.3.1. The relationship between PEAK and age. Pearson correlation coefficients were conducted to determine the relationship between PEAK total score, age, and sex. The primary relationship of interest was PEAK total score and age; however, the correlation between both of these variables and sex was assessed to determine if sex was a moderating factor. Also, a scatterplot was developed to further assess the relationship between PEAK and age for individuals with autism. These data were juxtaposed with the scatterplot data from the normative sample to determine the degree to which the autism sample deviated from the normative sample. An independent samples *t*-test was conducted between the autism sample and the normative sample across PEAK total scores to determine if the differences between the two groups were significant ($p < .05$).

3.1.3.2. Percentage distribution by PEAK score. A frequency distribution of the number of students across PEAK scores was created to determine the number of students at the school who could be appropriately served using the PEAK direct training curriculum. As Study 1 suggested, PEAK direct training may not be an appropriate tool for learners who have exceeded the language and cognitive abilities of a typically developing 8-year-old. Therefore, it is important to determine the utility of the PEAK direct training curriculum for a school-serving individuals with autism of varying ages. The participants' PEAK total scores were grouped by 23 (i.e. 0–23, 24–46, 57–69 ... 162–184), and the percentage of participants in each sample who fell within each PEAK total score range was assessed. A chi-square analysis was then conducted to determine if the difference between the groups was significant ($p < .05$).

3.2. Results

3.2.1. The relationship between PEAK and age

The results of the Pearson correlational analysis suggest that there were no significant relationships between PEAK total score and age ($r = .008$, $p = .947$), PEAK total score and gender ($r = .021$, $p = .861$), or age and gender ($r = .156$, $p = .197$). As the relationship between PEAK total score and age was the variable of interest, the results of this analysis are displayed in Fig. 4. Visual analysis of the data support the conclusion that there was no relationship between PEAK total score and age. Moreover, visual analysis suggests that although there was a ceiling effect for the normative sample around the

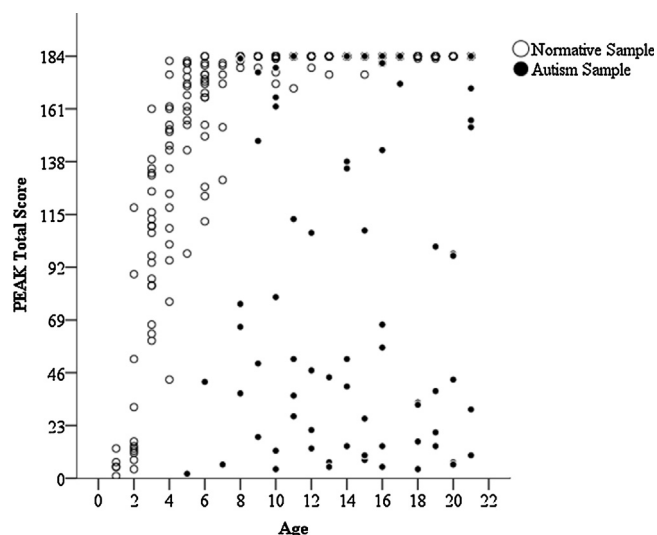


Fig. 4. The relationship between PEAK total score and Age for the autism sample and the normative sample.

8-years-of-age, the same ceiling effect was not observed in the autism sample. The results of the independent samples *t*-test suggest that the difference between the two groups was significant ($t(275) = 10.63, p < .001$).

3.2.2. Percentage distribution by PEAK score

The percentage distribution by PEAK total score is summarized in Fig. 5. The data suggest that the greatest percentage of participants at the school had PEAK total scores between 1 and 23 (30.4%). Conversely, the second greatest percentage of students at the school had PEAK total scores between 162 and 184 (20.2%). The smallest percentage of students at the school had PEAK total scores between 70 and 92 (2.8%). The data also suggest that most participants in the normative sample had PEAK total score between 162 and 184 (70.6%), and only 29.4% participants from the normative sample had PEAK total scores below 162. The frequency distribution of the autism group and the normative sample across PEAK total score groups was statistically significant ($\chi^2(7) = 86.9, p < .001$).

3.3. Discussion

The present data serve as a comparison of participants with autism to the normative sample. The results of the present study expand on psychometric evaluations of PEAK (Dixon, Carman, et al., in press-a; Dixon, Whiting, et al., in press-b), as well as extend the field's understanding of the nature of autism spectrum disorders. The data are not intended to serve as a

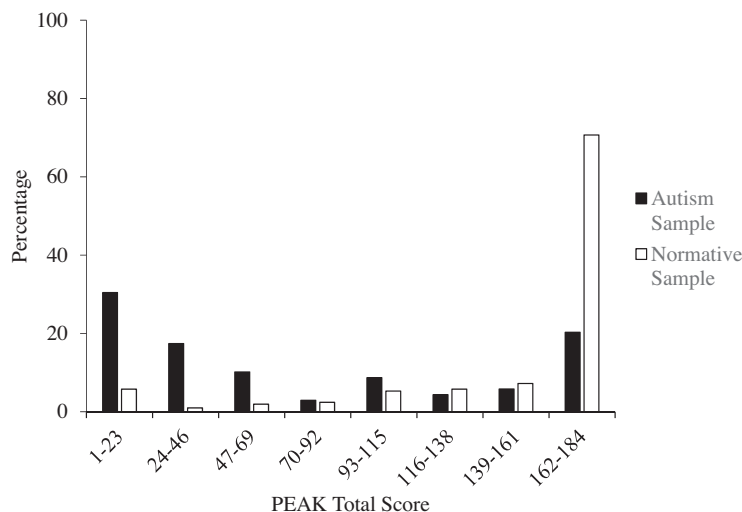


Fig. 5. Percentage distribution of participants in the autism sample and the normative sample across PEAK total scores.

representative sample of all individuals with autism, but rather to serve as an exemplar of how individuals can be compared to the normative sample to determine the utility of the PEAK direct training module with this population. The results of the study demonstrate that directly trained skill acquisition does not appear to develop in individuals with autism in the same manner as typically developing individuals. In fact, there was not a significant relationship between the age of the participants and their PEAK total scores. If individuals with autism have a directly trained skill repertoire that cannot be predicted by their age, then factors other than age must be more pertinent to language development (e.g., severity of the disability or access to vocational services).

The results also suggest that the directly trained skill repertoire of the autism sample was significantly lower than the normative sample. Although this result is not altogether surprising, the present data also suggest that most individuals from the autism sample could be adequately assessed and taught using the PEAK direct training curriculum. Only 20.2% of the participants had PEAK total scores greater than 162; the remaining 79.8% of the participants have more than 23 items that need to be mastered before continuing on to the remaining modules. In fact, the greatest frequency of participants fell between PEAK total scores of 1 and 23, representing completion of .5% and 12.5% of the skills on the assessment. The percentage distribution data suggest that individuals with autism may be better serviced using the PEAK assessment and curriculum than typically developing individuals, as 70.6% of the normative sample had PEAK total scores exceeding 162. Taken together, the data suggest that the directly trained skill repertoires of individuals with autism are different from that of typically developing individuals. By examining the source of these differences, strides can be made to improve the skills of individuals with autism, and begin to promote a sense of autonomy and freedom.

4. General discussion

The purpose of the present study was to develop a normative sample of the PEAK direct training module, and to compare a sample of participants with autism to the normative sample. The normative sample suggests that language and cognitive development, at least in terms of directly trained skills, may not be linear. Acquisition of directly trained material occurs rapidly early in life, and appears to slow down as children reach 8-years of age. At around the age of 8, PEAK direct training module becomes a psychometrically inaccurate measure of language and cognition from a behavior analytic perspective. Although this ceiling effect was apparent in the normative sample, the same ceiling effect was not apparent in the autism sample. In fact, there was no observed relationship between PEAK total score and age. As such, the data suggest that PEAK Direct Training may be an appropriate tool for individuals with autism across all chronological ages, contingent on their language and cognitive functioning level rather than their age. As PEAK has been shown to correlate with other measures of language and intelligence, it seems natural that the measure could be used as an assessment mechanism to pinpoint gaps in participants' language and cognitive development, and provide a curriculum to fill these gaps.

The present study had several limitations that limit the inferences that can be generated from the data. First, relative to normative samples for predicting intelligence (e.g., Wechsler, 2003), the present study had a relatively small sample size. However, it should be kept in mind that PEAK was not designed as an intelligence test to compare a single individual to a larger group norm. Instead the main purpose of PEAK has been noted to be an assessment and curriculum guide. Nonetheless, the relatively smaller sample of the present study far exceeds in size and psychometric sophistication what other behavior analytic attempts at normalization have been documented in the literature. Furthermore, it is important to note that significant results were observed with a smaller sample size, suggesting high internal validity, and larger sample sizes may be unnecessarily redundant. A second limitation was the relative homogeneity of the sample, which could be improved upon by subsequent investigations using a demographically diverse population that matches the distribution of persons with autism.

Future research could extend the results of the present study in several ways. As the remaining PEAK modules (equivalence and transformation) are published and validated, normative samples must be developed for these assessments. Not only will evaluating these assessments allow for a greater number of items for comparison to children with autism, but it will also allow for a better understanding of human language and cognitive development beyond the point of a typically developing 8-year-old. In addition, future research must show how behavior analytic interventions based on the PEAK program actually change pre-post intervention. To date we are not in a position to clearly state that a PEAK score for a given individual will in fact change following intervention. Conceptually, this change will occur given that PEAK: Direct Training has been explained as traditional discrete trial training/ABA therapy. However, no published data currently document group or individual child performances over time and how mathematical relationships such as the current correlational models are altered following intervention. In summary, the PEAK Relational Training System holds promise as an assessment for children with autism when clinicians are attempting to determine repertoire strengths and weaknesses. The current normalization data now also provide those clinicians with typical developmental age norms to assist in benchmarking targeted performances and discussions with parents and caregivers as to how a child with autism may deviate from typical developing peer of the same age.

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